

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ405			
Course Title	AXIOMATIC FOUNDATIONS OF MATHEMATICS			
Type of Course	Elective			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Nil			
Course Summary	The course goes into the philosophy of mathematics, modern axiom methods, controversies in set theory around axiom of choice, its implications and various philosophical alternative approaches to the foundations of mathematics.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse Axiomatic Systems and Logical Deductions	An	C	Internal Exam/Assignment / Seminar/ Viva / End Sem Exam
CO2	Explore Axioms and their Interpretation of Mathematical Structures	Ap	C	Internal Exam/Assignment / Seminar/ Viva / End Sem Exam
CO3	Investigate Properties of standard sets in Mathematics and obtain their axiomatic constructions	E	P	Internal Exam/Assignment / Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

TEXT: R. Wilder, Introduction to the Foundations of Mathematics (2/e), John Wiley & Sons, 1967				
Module	Unit	Content	Hrs (60)	Ext. Marks (70)
I	Axiomatic Method (Up to Chapter 3 Section 5 of Text Book)		12	
	1	Description - undefined terms, axioms, logical deductions and proofs. Case study with axioms of points and lines.		
	2	Axioms and Interpretation (models): consistency (satisfiability), completeness, categorically and independence.		
	3	Case Study with axioms of order and equivalence.		
	4	Sets and Russal's Paradox.		
	5	Finite and Infinite Sets,		
	6	Review of Mathematical Induction.		
II	Set Theory: Cardinals (Chapter 3, Section 6 to Chapter 4 of Text Book)		12	
	7	Infinite Sets - Ordinary and Dedekind Infinity and their equivalence		
	8	Axiom of Choice		
	9	Countable Sets and their properties		
	10	Diagonalization and Uncountable Sets, Irrational Numbers		
	11	Cardinal Numbers and Bernstein's Equivalence Theorem		
	12	Well Ordered Sets and Transfinite Induction		
III	Set Theory: Ordering (Chapter 5)		12	
	13	Well Ordering Theorem		
	14	Ordinals and Burali-Forti Paradox		
	15	Properties of Ordinals and Continuum Hypothesis		
	16	Equivalence of Axiom of Choice, Well Ordering Theorem.		
	17	Zorn's Lemma and Equivalence with Axiom of Choice		
IV	Real Numbers (Chapter 6 of Text Book)		12	
	18	Ordering and Separability of Reals, and Dedekind Cuts.		

	19	Axiomatization of Real Numbers: Constituency, Independence and		
	20	Categoricalness of Real Number Axioms.		
	21	Definition of Real numbers from Peano's Axioms		
	22	Complex Numbers.		
V	Discussions in Mathematical Philosophy			
	1	Abstractions: Groups/Rings/Fields/Vector Spaces		
	2	Zermelo Fraenkel Axiomatization of Set Theory		
	3	Frege-Russell Thesis Set Theory using Predicate Calculus		
	4	Brower's Intuitionist Theory		
	5	Formal Deductions and Godel's Theorems.		

References:

1. I. M. Copi, Symbolic Logic (5/e), Pearson, 2015.
2. U. C. Merzbach and C. B. Boyer, A History of Mathematics, (3/e), 2011.
3. I. Stewart and D. Tall, The foundations of Mathematics, (2/e), Oxford University Press 2015.

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	2	1	3	3	3	3	0	0	3
CO 2	3	3	2	1	3	3	3	3	0	0	3
CO 3	3	3	2	1	3	3	3	3	0	0	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓